

# WASP-12b

R-0376

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-12\_230222 · created 2026-07-18T18:22:15+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2459997.7251 +/- 0.001 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.153 +/- 0.045                |
| Transit depth (Rp/Rs)^2                           | 2.35 +/- 1.38 [%]              |
| Orbital Inclination (inc)                         | 86.76 +/- 2.89                 |
| Airmass coefficient 1 (a1)                        | 0.7 +/- 0.42                   |
| Airmass coefficient 2 (a2)                        | 0.24 +/- 0.71                  |
| Scatter in the residuals of the lightcurve fit is | 56.73 %                        |
| Best Comparison Star                              | None                           |
| Optimal Aperture                                  | 2.17                           |
| Optimal Annulus                                   | 3.5                            |
| Transit Duration (day)                            | 0.1328 +/- 0.0065              |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.153 ± 0.045 vs 0.1178 (deviation 0.78σ) |
| Duration fit vs published | 0.1328 d vs 0.125 d (deviation 1.2σ)      |
| Mid-time O–C              | -0.3 min                                  |
| Depth SNR                 | 3.4                                       |
| Residual scatter          | 56.73 %                                   |

## Light curve

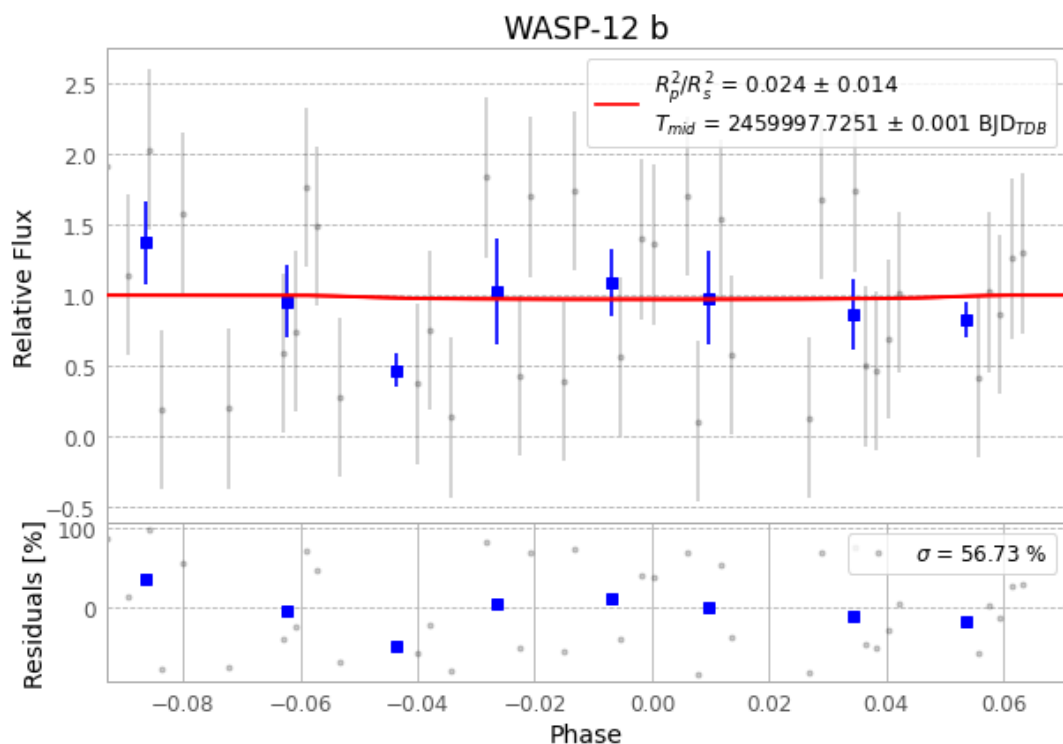


Figure 1 — FinalLightCurve\_WASP-12 b\_2023-02-21.png (SHA-256 21ac11a98955cca2...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [214, 108], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 553a19f5a00c84e6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 3007197

## Data provenance

87 FITS frames (57 MB), WASP-12230222025146.FITS → WASP-12230222071216.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-230222.log (27226bfa4bbe...), FinalLightCurve\_WASP-12 b\_2023-02-21.png (21ac11a98955...), FinalLightCurve\_WASP-12 b\_2023-02-21.csv (aea276d022cc...)

## Compute resources

Wall time 150.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 18:22Z.